

CENTRAL ILLINOIS, IL, USA

WMO: 724397

Lat: 40.479N

Lon: 88.915W

Elev: 264

StdP: 98.20

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 54831

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.3	-15.7	-22.6	0.5	-17.3	-20.1	0.6	-15.0	14.4	-1.9	12.9	-0.1	4.8	310	0.583

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	33.0	24.3	31.9	23.8	30.5	23.0	26.2	30.9	25.2	29.6	24.3	28.6	5.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.9	20.6	29.2	23.9	19.4	28.1	23.0	18.3	26.8	82.7	30.8	78.5	29.8	74.6	28.6	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
12.4	11.1	10.0	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-22.5	35.1	3.8	1.8	-25.2	36.4	-27.5	37.4	-29.6	38.4	-32.4	39.7		
			-22.9	27.9	3.8	1.0	-25.7	28.6	-27.9	29.1	-30.0	29.7	-32.8	30.4		
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Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.1	-4.0	-2.3	4.6	11.3	17.2	22.3	23.9	22.9	19.6	12.5	5.2	-0.9		
	DBStd	11.12	6.72	6.40	6.41	5.42	4.98	3.36	3.08	2.88	4.22	5.40	5.50	5.80		
	HDD10.0	1559	433	346	191	48	4	0	0	0	1	35	162	339		
	HDD18.3	3229	691	577	427	219	83	7	2	2	34	196	394	596		
	CDD10.0	1958	1	2	24	86	228	369	430	399	289	111	18	2		
	CDD18.3	585	0	0	1	7	48	126	173	143	73	14	0	0		
	CDH23.3	5254	0	0	11	78	443	1141	1639	1184	642	114	1	0		
	CDH26.7	1782	0	0	1	13	128	395	613	388	220	24	0	0		
(14) Wind	WSAvg	5.0	5.8	5.6	5.8	5.9	5.2	4.5	3.8	3.5	3.9	5.0	5.6	5.5		
(15) Precipitation	PrecAvg	961	61	48	67	91	116	107	100	95	62	82	80	54		
	PrecMax	1335	164	138	136	183	248	219	307	157	193	157	166	153		
	PrecMin	831	19	9	26	38	19	28	40	35	18	31	9	29		
	PrecStd	139	32	34	30	36	62	52	63	35	42	40	38	31		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.2	17.3	25.0	28.3	31.9	33.8	36.2	34.2	33.8	29.4	21.7	16.1		
		MCWB	13.8	13.6	17.6	18.3	21.9	23.1	24.5	25.1	22.9	21.4	15.1	14.7		
	2%	DB	11.3	13.1	20.4	25.8	29.8	32.3	33.0	32.3	31.3	26.4	18.3	12.6		
		MCWB	9.8	10.4	15.5	17.0	20.3	23.1	25.4	24.5	22.5	18.9	14.4	11.1		
	5%	DB	8.0	9.4	17.5	23.0	28.0	30.9	31.7	30.5	29.3	24.0	16.1	9.8		
		MCWB	6.6	7.0	13.4	15.7	19.8	22.7	24.7	23.5	21.2	17.9	13.1	8.2		
	10%	DB	4.9	6.5	14.6	20.7	25.9	29.0	30.1	28.8	27.3	21.2	13.7	7.3		
		MCWB	3.5	4.3	11.3	14.4	18.9	21.8	23.6	22.6	19.9	16.5	10.9	5.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.3	14.4	18.3	20.1	23.6	26.1	27.9	27.0	25.0	22.7	17.4	14.7		
		MCDB	15.3	15.8	23.1	25.8	29.2	30.9	32.5	31.6	30.7	27.8	19.2	15.5		
	2%	WB	10.2	10.7	16.3	18.4	22.3	25.0	26.6	25.8	23.8	20.9	15.4	11.4		
		MCDB	11.1	12.8	19.3	23.1	27.5	29.9	31.2	30.4	29.1	24.6	17.4	12.1		
	5%	WB	6.6	7.5	14.4	16.8	21.1	24.0	25.6	24.8	22.7	19.2	13.5	8.2		
		MCDB	7.9	9.1	17.3	21.5	25.7	28.4	29.8	28.9	27.1	21.8	15.7	9.3		
	10%	WB	3.5	4.5	11.7	15.4	20.0	23.0	24.6	23.8	21.6	17.5	11.2	5.6		
		MCDB	4.7	6.4	14.2	19.9	24.2	27.6	28.7	27.5	25.4	20.7	13.6	6.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	8.9	10.1	11.5	11.3	10.7	10.3	10.4	11.7	10.6	9.3	7.5		
		MCDBR	11.1	12.5	13.5	14.6	13.2	12.3	11.2	11.9	13.4	13.5	12.3	10.5		
	5% WB	MCWBR	9.7	9.8	8.8	8.2	6.3	5.6	5.3	5.6	5.5	7.0	8.9	9.5		
		MCWBR	10.8	12.0	12.3	12.8	11.7	10.2	10.1	10.2	11.2	10.9	11.3	10.0		
(40) Clear-Sky Solar Irradiance	taub		0.294	0.307	0.344	0.387	0.426	0.438	0.452	0.443	0.403	0.353	0.309	0.292		
	taud		2.437	2.398	2.383	2.269	2.213	2.224	2.169	2.182	2.270	2.415	2.493	2.459		
	Ebn at Noon		870	909	907	886	854	842	825	822	832	844	848	844		
	Edh at Noon		79	95	108	130	141	140	146	141	121	93	75	72		
(44) All-Sky Solar Radiation	RadAvg		1.80	2.70	3.67	4.75	5.46	6.22	6.22	5.46	4.64	3.13	2.08	1.52		
	RadStd		0.21	0.25	0.28	0.38	0.44	0.42	0.32	0.34	0.35	0.35	0.18	0.19		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+84	N/A

Nomenclature: See separate page